

Dynamic Programming

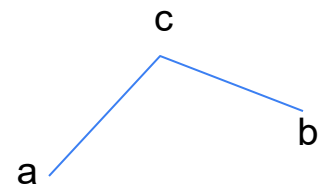


Dynamic Programming

High-level Idea: Break a problem into smaller subproblems (like divide-and-conquer) but divide into **many overlapping** subproblems.

The DP technique applies to problems that obey the **principle of “optimal substructure”**: the overall optimal solution can be constructed from optimal solutions to smaller subproblems

Example: A subpath of any shortest path is itself a shortest path.



Warm-Up: Fibonacci

Input: non-negative integer n

Output: n^{th} Fibonacci number

Recurrence for Fibonacci: $F(n) = F(n-1) + F(n-2)$

$$F(0) = F(1) = 1$$



Given a recurrence, three ways to compute its values:

1. **Top-down Recursive:** Starting at desired input, recurse down to base case(s)
2. **Top-down with Memoization:** Same as 1, but save results as they're computed, reusing already-computed results
3. **Bottom-Up Table Filling (aka Dynamic Programming):** Start from base case(s), build up to desired result

Fibonacci Method 1: Top-down Recursive

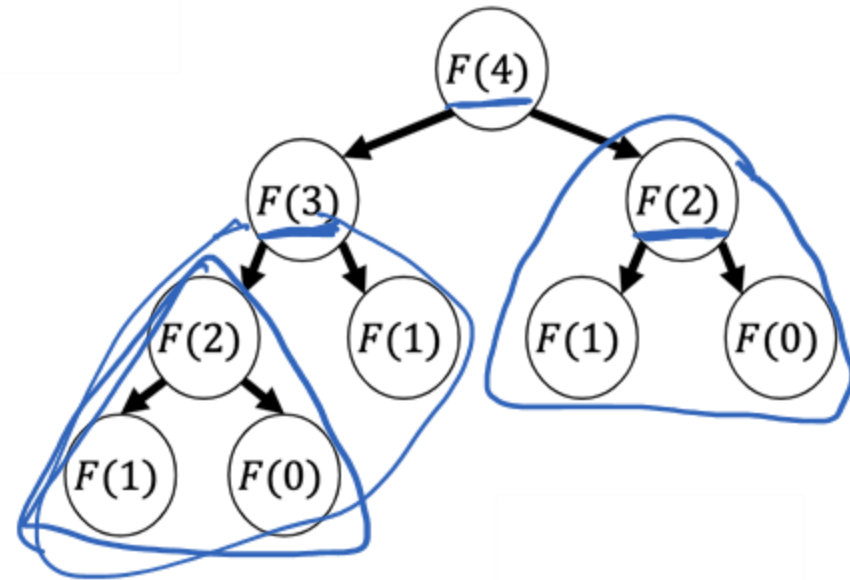
Algorithm Fib(n):

If $n = 0$ OR $n = 1$:

Return 1

Else:

Return Fib(n-1) + Fib(n-2)



- **Pro:** direct translation of recurrence into code
- **Con:** $\exp(n)$ recursive calls

Fibonacci Method 2: Top-down with Memoization

memo := array indexed from 0 to n

Algorithm Fib(n):

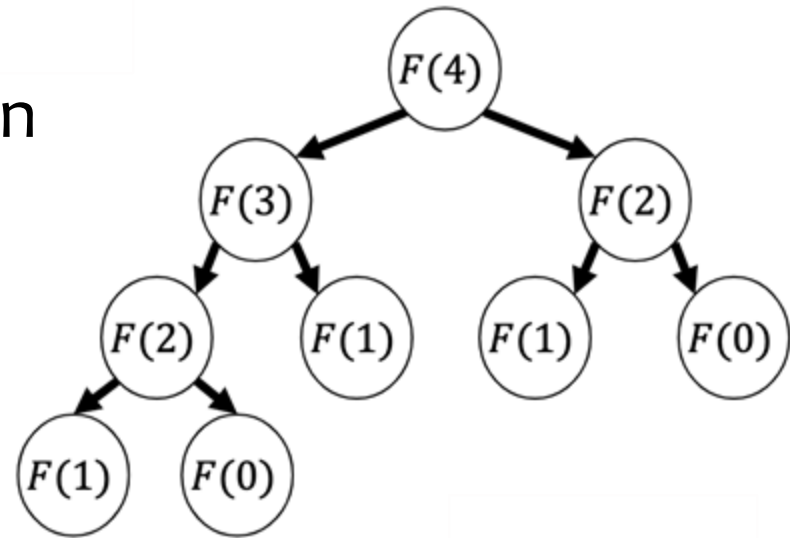
If $n = 0$ OR $n = 1$:

Return 1

Else if **memo**(n) is unoccupied:

memo(n) = Fib(n-1) + Fib(n-2)

Return memo(n)



- **Pro:** way faster than method 1
- **Con:** not clear how to analyze running time

Fibonacci Method 3: Bottom-Up Table Filling (aka Dynamic Programming)

Algorithm Fib(n):

table := array indexed from 0 to n

table(0) = 1

table(1) = 1

for i = 2 to n:

table(i) = **table**(i-1) + **table**(i-2)

Return table(n)

This still isn't poly time since the input size is $\log n$, but this **method** gives poly time algorithms for many other problems



- **Pros:** gives roadmap for running time analysis:
(size of table) $\cdot$ (time to fill each entry)
- **Con:** need to translate from recurrence to filling table bottom-up



First use of “Fibonacci” numbers:
Pingala: mathematician and poet
India ~200 BCE



Ferret's Library
@ferretslibrary

Subscribe



FUN FACT

Fibonacci is a made up name for Leonardo Bonacci of Pisa. The name Fibonacci comes from the Latin "*filius Bonacci*", "the son of Bonacci". His father was Guglielmo Bonaccio.

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...



9:22 AM · Jan 31, 2024 · 52K Views



Now let's solve a problem using Dynamic Programming!

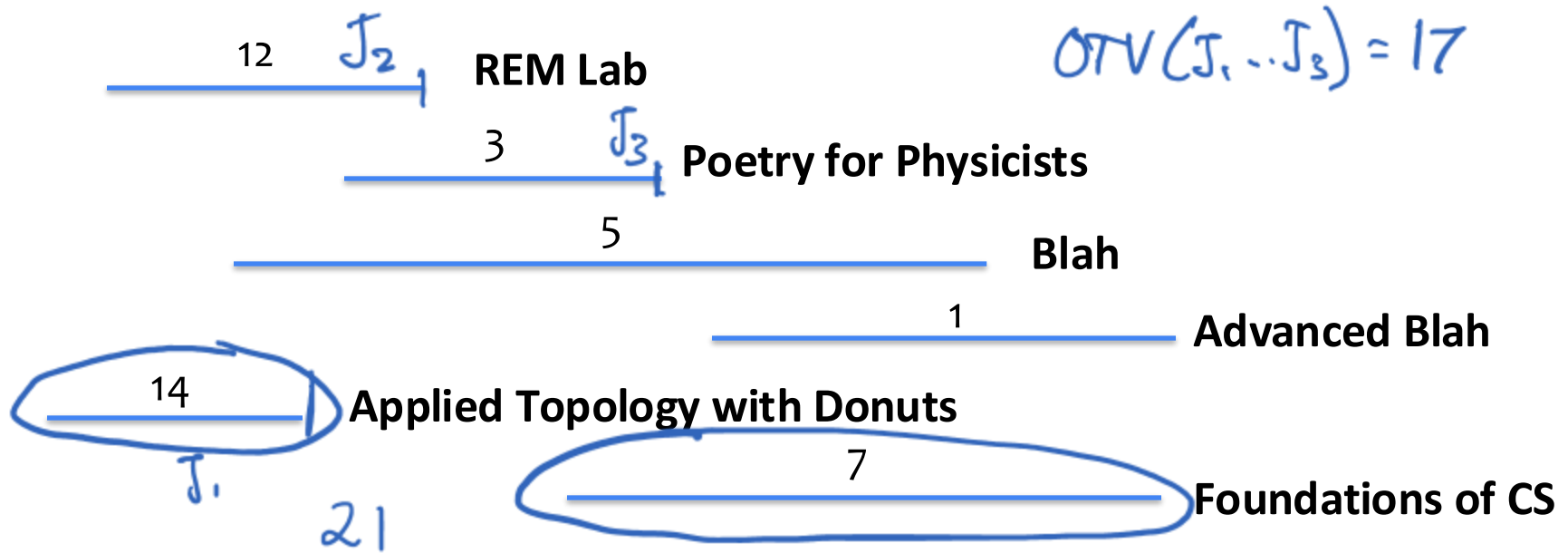
What we'll do:

1. Derive a recurrence to express the solution value
2. Compute the values of the recurrence using method 3 (bottom-up table filling).
3. Construct a solution from the recurrence values

*Method 2 (top-down with memoization) is sometimes also considered “dynamic programming” but we'll **only** focus on method 3 (bottom-up table filling) since it helps us calculate the running time.

Weighted Course Registration Problem

(aka Weighted Task Selection)



Goal: Choose a set of non-intersecting courses with largest total value.
(there may be many optimal solutions, we just seek one)

I'd take the donuts course if it didn't create a hole in my schedule!



*Let the input size be $n = \text{\#intervals}$. (Assume the weights are small enough that we can do arithmetic operations on them in constant time.)

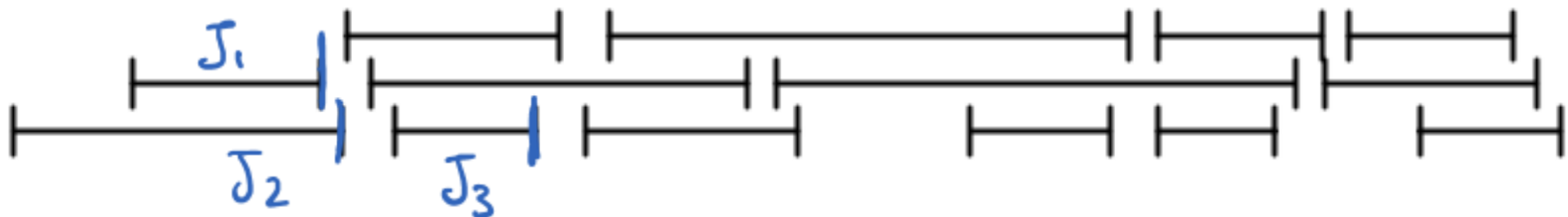
Weighted Task Selection Recurrence

Assume the intervals $J_1, J_2, \dots, J_n$ are sorted by **finish time**.

Let $\mathbf{OPT}(J_1, \dots, J_i)$ be an optimal solution for $J_1, \dots, J_i$.

Let $\mathbf{OTV}(J_1, \dots, J_i)$ be the **value** of the optimal solution for $J_1, \dots, J_i$.

“Optimal Task Value”



Weighted Task Selection Recurrence

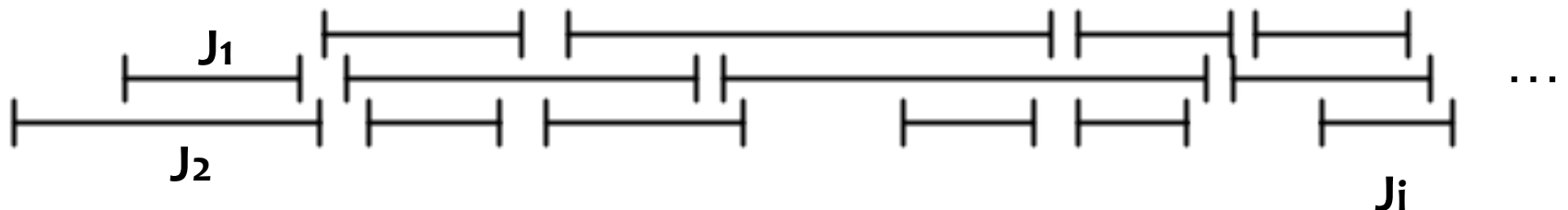
Assume the intervals $J_1, J_2, \dots, J_n$ are sorted by **finish time**.

Let $\text{OPT}(J_1, \dots, J_i)$ be an optimal solution for $J_1, \dots, J_i$.

Let $\text{OTV}(J_1, \dots, J_i)$ be the **value** of the optimal solution for $J_1, \dots, J_i$.
“Optimal Task Value”

There are two options:

- $\text{OPT}(J_1, \dots, J_i)$ has J_i **(use it!)**
- $\text{OPT}(J_1, \dots, J_i)$ doesn't have J_i **(lose it!)**



Weighted Task Selection Recurrence

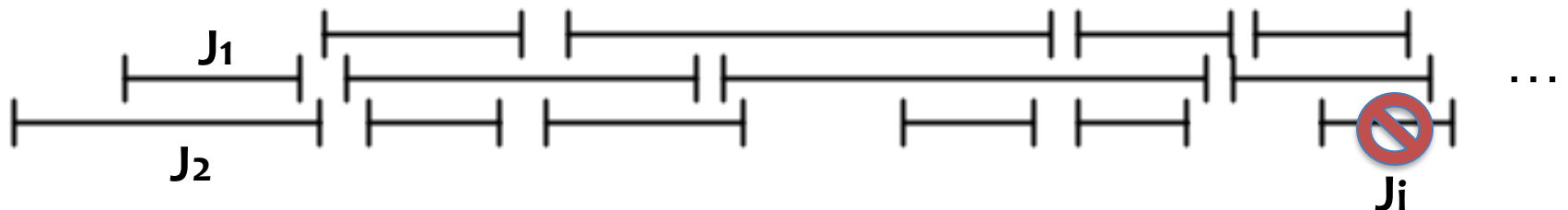
Assume the intervals $J_1, J_2, \dots, J_n$ are sorted by **finish time**.

Let $\text{OPT}(J_1, \dots, J_i)$ be an optimal solution for $J_1, \dots, J_i$.

Let $\text{OTV}(J_1, \dots, J_i)$ be the **value** of the optimal solution for $J_1, \dots, J_i$.
“Optimal Task Value”

There are two options:

- $\text{OPT}(J_1, \dots, J_i)$ has J_i (**use it!**)
- $\text{OPT}(J_1, \dots, J_i)$ doesn't have J_i (**lose it!**) $\Rightarrow \text{OTV}(J_1, \dots, J_i) = \text{OTV}(J_1, \dots, J_{i-1})$



Weighted Task Selection Recurrence

Assume the intervals $J_1, J_2, \dots, J_n$ are sorted by **finish time**.

Let $\text{OPT}(J_1, \dots, J_i)$ be an optimal solution for $J_1, \dots, J_i$.

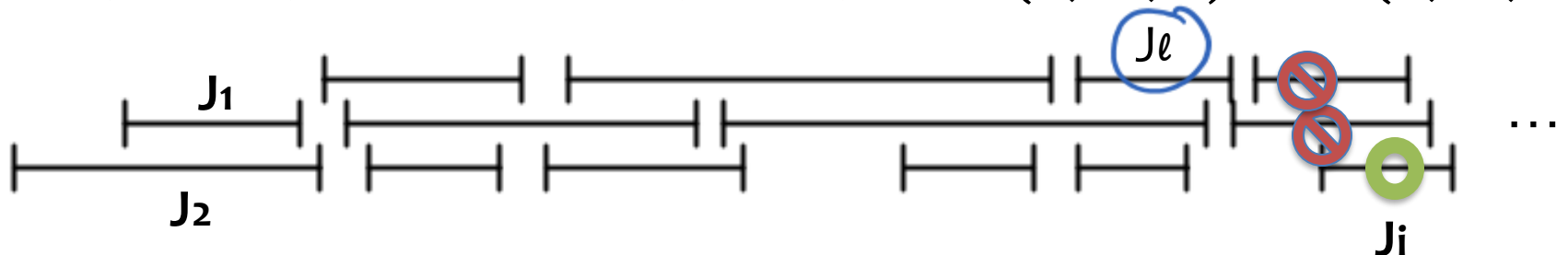
Let $\text{OTV}(J_1, \dots, J_i)$ be the **value** of the optimal solution for $J_1, \dots, J_i$.
“Optimal Task Value”

There are two options:

- $\text{OPT}(J_1, \dots, J_i)$ has J_i (**use it!**) $\Rightarrow \text{OTV}(J_1, \dots, J_i) = \text{val}(J_i) + \text{OTV}(J_1, \dots, J_\ell)$

where J_ℓ is the **last** interval (before J_i) that doesn't overlap with J_i

- $\text{OPT}(J_1, \dots, J_i)$ doesn't have J_i (**lose it!**) $\Rightarrow \text{OTV}(J_1, \dots, J_i) = \text{OTV}(J_1, \dots, J_{i-1})$



The Final Recurrence

$$OTV(I_1, \dots, I_n) = \max \left\{ \text{val}(I_n) + OTV(I_1, \dots, I_i), \right. \\ \left. OTV(I_1, \dots, I_{n-1}) \right\}$$

$$OTV(\emptyset) = 0$$

You could write code to directly implement this recurrence as an algorithm...

Algorithm OTV($J_1, \dots, J_n$):

if $n = 0$:

Return 0

else:

ℓ = index of last interval (before J_n) that doesn't overlap with J_n

Return $\max\{\text{val}(J_n) + \text{OTV}(J_1, \dots, J_\ell), \text{OTV}(J_1, \dots, J_{n-1})\}$

... but it would run in exponential time

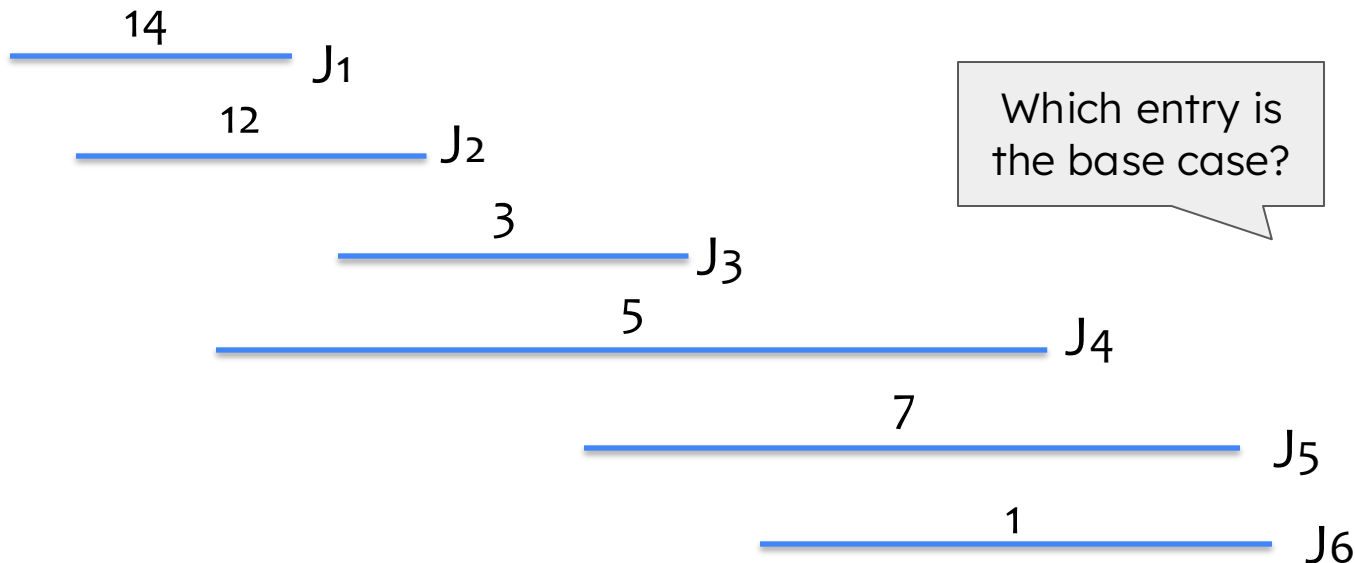
Dynamic Programming in Action!

(use it)

(lose it)

For reference: $\text{OTV}(J_1, \dots, J_i) = \max\{\text{val}(J_i) + \text{OTV}(J_1, \dots, J_{i-1}), \text{OTV}(J_1, \dots, J_{i-1})\}$

$\text{OTV}(\emptyset)$	$\text{OTV}(J_1)$	$\text{OTV}(J_1, J_2)$	$\text{OTV}(J_1, \dots, J_3)$	$\text{OTV}(J_1, \dots, J_4)$	$\text{OTV}(J_1, \dots, J_5)$	$\text{OTV}(J_1, \dots, J_6)$
0						



Which entry is the base case?

Which cell contains the final answer?

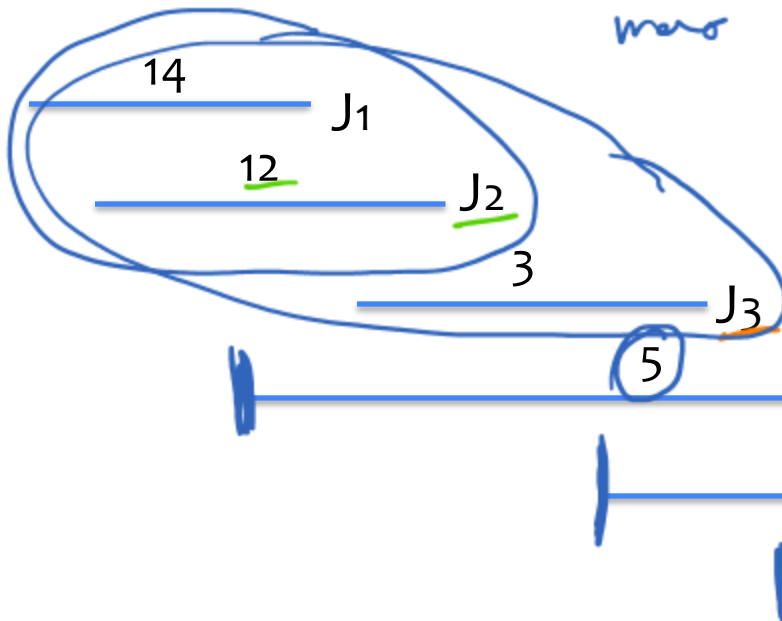


Dynamic Programming in Action!

For reference: $\text{OTV}(J_1, \dots, J_i) = \max\{\text{val}(J_i) + \text{OTV}(J_1, \dots, J_{i-1}), \text{OTV}(J_1, \dots, J_{i-1})\}$

$\text{OTV}(\emptyset)$	$\text{OTV}(J_1)$	$\text{OTV}(J_1, J_2)$	$\text{OTV}(J_1, \dots, J_3)$	$\text{OTV}(J_1, \dots, J_4)$	$\text{OTV}(J_1, \dots, J_5)$	$\text{OTV}(J_1, \dots, J_6)$
0	14	14	17	17	21	21

Handwritten annotations on the table: Above $\text{OTV}(J_1)$ is 14 (orange) with a blue arrow pointing to the cell. Above $\text{OTV}(J_1, J_2)$ is 14 (orange) with a blue arrow pointing to the cell. Above $\text{OTV}(J_1, \dots, J_3)$ is 17 (orange) with a blue arrow pointing to the cell. Above $\text{OTV}(J_1, \dots, J_4)$ is 17 (orange) with a blue arrow pointing to the cell. Above $\text{OTV}(J_1, \dots, J_5)$ is 21 (orange) with a blue arrow pointing to the cell. Above $\text{OTV}(J_1, \dots, J_6)$ is 21 (orange) with a blue arrow pointing to the cell. Below the table, green arrows labeled "use" point from $\text{OTV}(J_1)$ to $\text{OTV}(J_1, J_2)$, from $\text{OTV}(J_1, J_2)$ to $\text{OTV}(J_1, \dots, J_3)$, from $\text{OTV}(J_1, \dots, J_3)$ to $\text{OTV}(J_1, \dots, J_4)$, from $\text{OTV}(J_1, \dots, J_4)$ to $\text{OTV}(J_1, \dots, J_5)$, and from $\text{OTV}(J_1, \dots, J_5)$ to $\text{OTV}(J_1, \dots, J_6)$. Red arrows labeled "lose" point from $\text{OTV}(J_1)$ to $\text{OTV}(J_1, J_2)$, from $\text{OTV}(J_1, J_2)$ to $\text{OTV}(J_1, \dots, J_3)$, from $\text{OTV}(J_1, \dots, J_3)$ to $\text{OTV}(J_1, \dots, J_4)$, from $\text{OTV}(J_1, \dots, J_4)$ to $\text{OTV}(J_1, \dots, J_5)$, and from $\text{OTV}(J_1, \dots, J_5)$ to $\text{OTV}(J_1, \dots, J_6)$. Below the table, the expression $5 + 0, 17$ is written in blue.



To fill in a cell, which other cells do we need to look at?

In which order do we fill the table?



Dynamic Programming in Action!

(use it)

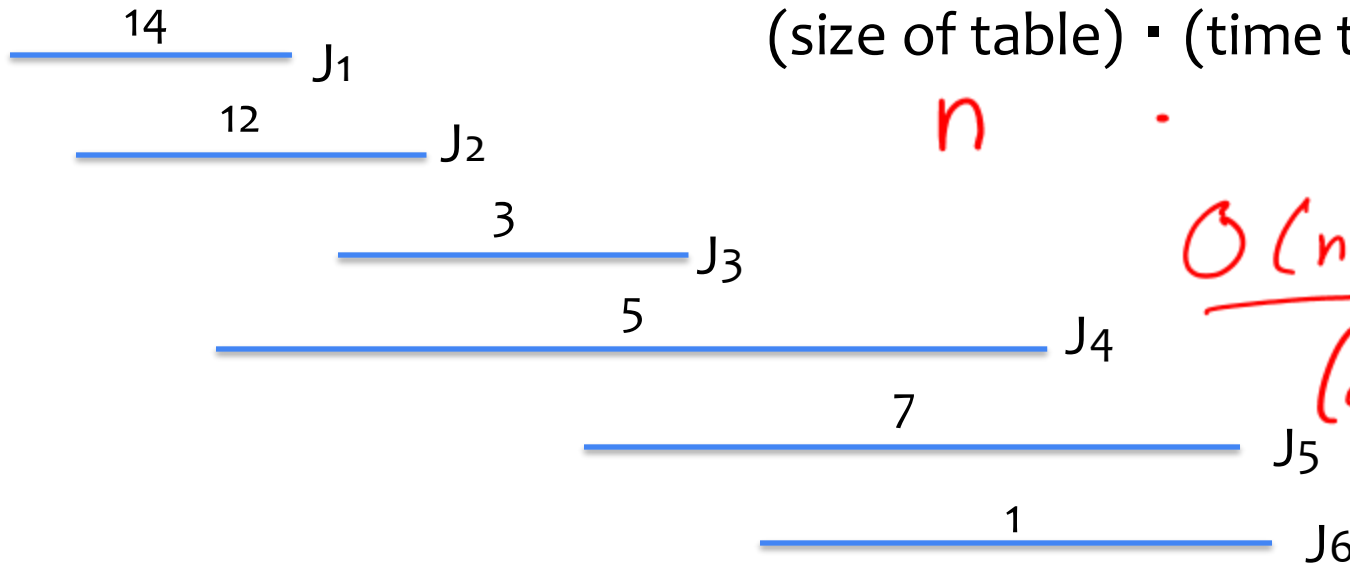
(lose it)

For reference: $\text{OTV}(J_1, \dots, J_i) = \max\{\text{val}(J_i) + \text{OTV}(J_1, \dots, J_{i-1}), \text{OTV}(J_1, \dots, J_{i-1})\}$

OTV($\emptyset$)	OTV(J_1)	OTV(J_1, J_2)	OTV($J_1, \dots, J_3$)	OTV($J_1, \dots, J_4$)	OTV($J_1, \dots, J_5$)	OTV($J_1, \dots, J_6$)

Running time =

(size of table) $\cdot$ (time to fill each entry)



$n \cdot n$

$O(n^2)$

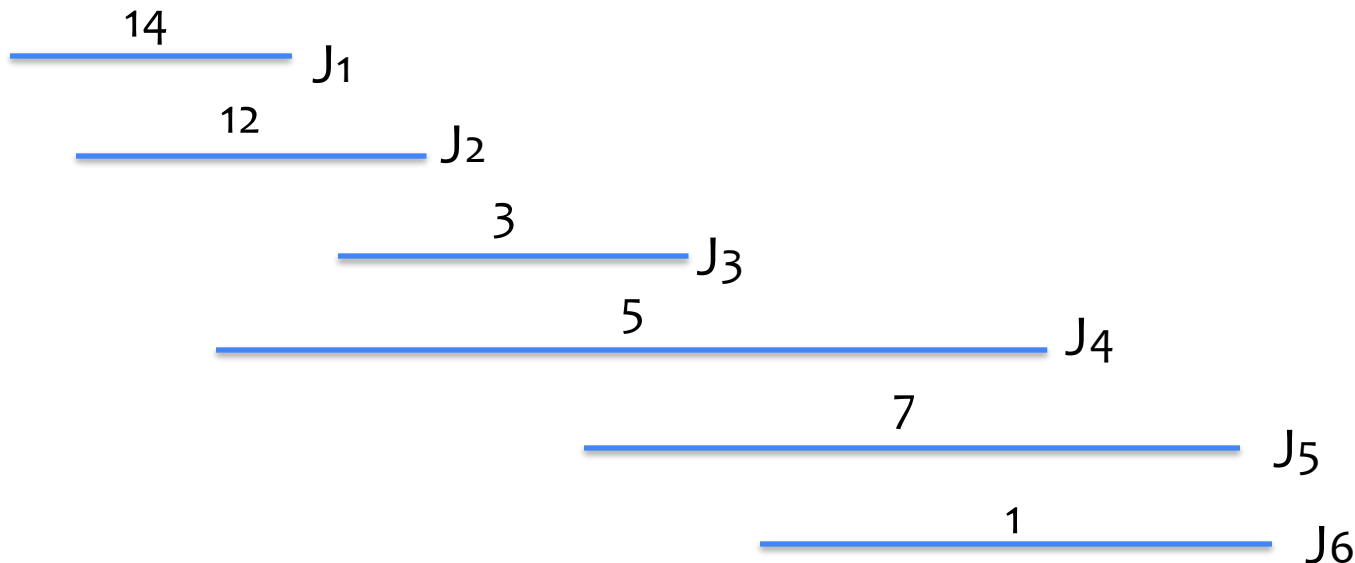
(actually $O(n \log n)$)

Dynamic Programming in Action!

For reference: $\text{OTV}(J_1, \dots, J_i) = \max_{\substack{\text{(use it)} \\ \text{(lose it)}}} \{ \text{val}(J_i) + \text{OTV}(J_1, \dots, J_{i-1}), \text{OTV}(J_1, \dots, J_{i-1}) \}$

$\text{OTV}(\emptyset)$	$\text{OTV}(J_1)$	$\text{OTV}(J_1, J_2)$	$\text{OTV}(J_1, \dots, J_3)$	$\text{OTV}(J_1, \dots, J_4)$	$\text{OTV}(J_1, \dots, J_5)$	$\text{OTV}(J_1, \dots, J_6)$

Reconstructing a solution



The Pseudocode

Algorithm OTV($J_1, \dots, J_n$):

table := array indexed from 0 to n

table(0) = 0

for $i = 1$ to n :

ℓ = index of last interval (before J_i) that doesn't overlap with J_i

table(i) = $\max\{\text{val}(J_i) + \text{table}(\ell), \text{table}(i-1)\}$

Return **table**(n)

The DP Recipe

“Let $OTV(J_1, \dots, J_i)$ be...”

1. Define *value version* of what we want to compute
2. Derive a recurrence ←

usually the trickiest part
3. Size of table: How many dimensions? Range of each dimension?
4. What are the base case(s)?
5. To fill in a cell, which other cells need to be filled already? In which order do I fill the table?
6. Which cell(s) contain the final answer?
7. Running time is often (size of table) $\cdot$ (time to fill each entry)
8. To reconstruct the solution (instead of just its value) follow arrows from final answer to base case

The Pseudocode

Algorithm OTV($J_1, \dots, J_n$):

table := array indexed from 0 to n $\leftarrow$ step 3 of DP recipe

table(0) = 0 $\leftarrow$ step 4 of DP recipe

for $i = 1$ to n : $\leftarrow$ step 5 of DP recipe

ℓ = index of last interval (before J_i) that doesn't overlap with J_i

table(i) = $\max\{\text{val}(J_i) + \text{table}(\ell), \text{table}(i-1)\}$

Return table(n) $\leftarrow$ step 6 of DP recipe $\uparrow$ step 2 of DP recipe

Structure of **all** DP pseudocode: loop through table in appropriate order and fill in cells according recurrence.

$\Rightarrow$ After answering DP Recipe questions, no creativity is needed!

Why is it called “dynamic programming”?

We had a very interesting gentleman in Washington named Wilson. He was Secretary of Defense, and he actually had a pathological fear and hatred of the word, research.

[...]

His face would suffuse, he would turn red, and he would get violent if people used the term, research, in his presence.

[...]

The RAND Corporation [where Bellman worked] was employed by the Air Force, and the Air Force had Wilson as its boss, essentially. Hence, **I felt I had to do something to shield Wilson and the Air Force from the fact that I was really doing mathematics inside the RAND Corporation.**

[...]

In the first place I was interested in planning, in decision making, in thinking. But planning, is not a good word for various reasons. I decided therefore to use the word, ‘programming.’

[...]

It’s impossible to use the word, dynamic, in a pejorative sense. Try thinking of some combination that will possibly give it a pejorative meaning. It’s impossible. Thus, I thought dynamic programming was a good name. It was something not even a Congressman could object to. So I used it as an umbrella for my activities.

-Richard Bellman (introduced dynamic programming in 1953)



Another example of DP:

Longest Increasing Subsequence (LIS)

Input: Array A of n numbers

Output: A longest increasing subsequence of A

Example: What is an LIS of [1,4,3,7,2]? *1 4 7, 1 3 7*

Let $LIS(A[1..i])$ be the length of the LIS of the subarray $A[1..i]$.

Recurrence for LIS

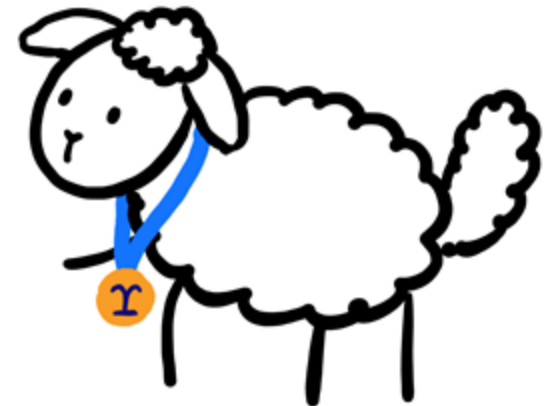
It's easy! Check it out!

$$\text{LIS}(A[1..i]) = \begin{cases} \text{LIS}(A[1..i-1]) + 1 & \text{if } A[i] > A[i-1] \quad \text{(use it!)} \\ \text{LIS}(A[1..i-1]) & \text{otherwise} \quad \text{(lose it!)} \end{cases}$$

Counterexample:

~~1 2 3 4~~

1 5 6 3 4



Recurrence for LIS

Defn: Let $\text{END-LIS}(A[1..i])$ be the length of an LIS of $A[1..i]$ *that ends in $A[i]$* .

When $A = [1, 3, 4, 2, \cancel{8}, 5]$, what is $\text{END-LIS}(A[1..i])$ for each i ?

1 2 (3) 2 4 4

1 +

Recurrence:

$\text{END-LIS}(A[1..i]) = 1 + \max\{\text{END-LIS}(A[1..j]) \text{ among all } j < i \text{ with } \underline{A[j] < A[i]}\}$
(if such j exists)

Base case: $\text{END-LIS}(A[1..i]) = 1$ if $A[i]$ is the smallest element so far in A

Let's follow the DP Recipe

$\text{END-LIS}(A[1..i]) = 1 + \max\{\text{END-LIS}(A[1..j]) \text{ among all } j < i \text{ with } A[j] < A[i] \text{ (if such } j \text{ exists)}\}$

A =	<u>3</u>	<u>10</u>	<u>1</u>	<u>4</u>	5	<u>9</u>	<u>2</u>	4
END-LIS(A[1..i])	<u>1</u>	2	<u>1</u>	<u>2</u>	3	<u>4</u>	2	3

base cases

final answer: max 4

Running time: $n \text{ cells} \cdot n \text{ time/cell}$ $O(n^2)$ time

Pseudocode

Algorithm LIS($A[1..n]$):

table := array indexed from 1 to n

for $i = 1$ to n :

if $A[i]$ is the minimum so far: // determined by scanning $A[1..i-1]$

table(i) = 1 // base case

Else:

table(i) = $1 + \max\{\mathbf{table}(j) \text{ among all } j < i \text{ with } \mathbf{A}[j] < \mathbf{A}[i]\}$

// find j by scanning **table** and **A**

Return $\max_i \mathbf{table}(i)$

Structure of **all** DP pseudocode: loop through table in appropriate order and fill in cells according recurrence.

⇒ After answering DP Recipe questions, no creativity is needed!

Something to try...

Try solving these problems by divide-and-conquer instead of DP and see how it doesn't seem to work.